

JS Crossdating Event Diagnosis Workflow

From format-preserving RWL data to one traceable event edit

STEP 1 DATA STATE AND REFERENCE CHRONOLOGY

Transparent RWL state

- Tucson short/long formats
- Raw baseline + working series
- Format-preserving read and write
- Target + same-site reference cores
- Year, zero and edit provenance

Reference chronology & side evidence

- Manual calendar-year chronology
- COFECHA-pass anchor master
- Spline detrending + AR prewhitening
- Leave-target-out reference
- PART 6 segment evidence
- Independent-core support

STEP 2 CONSTRAINED EVENT EVIDENCE

Lag topology

- Global sliding match
- COFECHA-like segment correlation
- Constrained integer-lag Viterbi path
- Direction, magnitude and plateaus
- Review windows: 5/7/9/13 years

Executable counterfactuals

- Virtual insert and delete
- Partial shifts: -2 to -100 years
- Whole-series shift grid
- Before/after correlations and residuals
- Executable operations only

Multi-core evidence ledger

- Master + independent-core votes
- Vote count, margin and concentration
- Terminal/fixed-side baseline
- Presence, operation and location types
- Provenance $\neq$ evidence weight

Missing ring

$$g - 1 \rightarrow g$$

False ring

$$g + 1 \rightarrow g$$

Partial shift

$$g - q \rightarrow g; q \geq 2$$

Whole-series shift

$$\text{constant global } g$$

STEP 3 JOINT ADJUDICATION: OPERATION $\times$ SHIFT $\times$ LOCATION

Linked hypothesis

Operation, lag path and evidence remain connected.

Joint adjudication

Fuse topology, counterfactuals and core votes.

Single product output

One executable edit + one focused review window.

Precise resolution

Resolve operation, shift and event location.

STEP 4 REVIEW, CONTROLLED COMMIT AND REDIAGNOSIS

Graph review

Inspect window and evidence

Edit preview

Select year or firstFixedYear

Confirm once

Apply one focused event

Refresh diagnosis

Rebuild on current working state

RwlEditor commit

Undo stack + operation log

FORMAT-PRESERVING

COUNTERFACTUAL

MULTI-CORE

FULLY AUDITABLE

updated working state